

LinkedIn Post:

https://www.linkedin.com/posts/janaashraf88_dotnet-csharp-entityframework-activity-7436191178460626944-L7si?utm_source=share&utm_medium=member_desktop&rcm=ACoAAE8YvOUBdlbw_Me3Yv_nmlqA8meTZp5UEhc

Self Study:

1. Association Relationship (Aggregation too) → has a/an

An **association relationship** shows a link between two entities where one class “**uses**” or “**knows about**” the other. It is often described as “**has a**”.

- **Example:** A Car **has an** Engine.

Aggregation is a **special type of association** where one entity **contains another**, but the contained entity can exist independently.

- **Example:** A Library **has many** Books. The books can exist even if the library is deleted.

2. Lazy vs Eager Loading

- **Lazy Loading** loads related data **only when it is accessed**.
 - Saves memory and performance if you don't always need the data.
 - **Example:** Accessing customer.Orders only when you call it.

- **Eager Loading** loads **all related data immediately** when the parent entity is loaded.
 - Ensures you have all data at once but may use more memory.
 - **Example:** Loading Orders along with Customer in the same query.
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3. One-to-One Relationship (Simple as One-to-Many)

A **one-to-one relationship** occurs when **one entity is linked to exactly one other entity**, similar to one-to-many but with a **maximum of one** on each side.

- **Example:** A User **has a** Profile. Each user has exactly one profile, and each profile belongs to exactly one user.
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4. Soft Delete

Soft Delete is a technique where a record is **not physically removed** from the database but is marked as deleted.

- Usually, a column like `IsDeleted` or `DeletedAt` is updated.
 - **Benefits:** You can recover deleted data, maintain history, and avoid accidental loss.
 - **Example:** Setting `IsDeleted = true` instead of deleting a Customer row.
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5. Restrict vs No Action

These are **referential integrity rules** in databases:

- **Restrict:** Prevents deleting or updating a parent row if child rows exist.
 - **No Action:** Does not automatically enforce restrictions; the database may throw an error if constraints are violated during a delete or update.
 - **Difference:** Both prevent invalid operations, but **Restrict actively blocks** the action immediately, while **No Action relies on deferred checking**.
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6. Index for Composite PK vs Column Primary Key

- **Column Primary Key:** A single column uniquely identifies each row. The database automatically creates an index on it.
 - **Composite Primary Key:** Two or more columns together **uniquely identify a row**. An index is usually created on the combination of these columns.
 - **Example:** OrderDetail table has OrderId + ProductId as a composite key, because either column alone is not unique.
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7. Self Relationship

A **self-relationship** occurs when an entity is **related to itself**.

- Often used for hierarchical data.

- **Example:** An Employee **reports to** another Employee.
- Another example: A Category **has a Parent Category** and may have **Child Categories**.